

Mar 24, 2026

$$\begin{aligned} \dot{x}_1 &= x_1^2 - x_1^3 + x_2 = f_1(x_1, \alpha) + z_2 \\ \dot{x}_2 &= x_3 = \phi + z_3 \\ \dot{x}_3 &= u \end{aligned}$$

$$\alpha(x_1) = -x_1^2$$

$$V(x_1) = \frac{1}{4} x_1^4$$

$$\dot{V}(x_1) = x_1^3 \dot{x}_1$$

$$= x_1^3 (x_1^2 - x_1^3 - x_1^2)$$

$$= -x_1^6$$

$$\alpha(x_1) = -x_1^2 + x_1^3 - x_1$$

$$V(x_1) = \frac{1}{2} x_1^2$$

$$z_2 = x_2 - \alpha(x_1)$$

$$\dot{z}_2 = \dot{x}_2 - \dot{\alpha}$$

$$V_2(x_1, z_2) = V_1(x_1) + \frac{1}{2} z_2^2$$

$$\dot{V}_2 = \dot{V}_1 + z_2 \dot{z}_2$$

$$= \frac{\partial V_1}{\partial x_1} [f_1(x_1, \alpha) + z_2] + z_2 [\phi(x_1, x_2) - \dot{\alpha}]$$

$$= \frac{\partial V_1}{\partial x_1} [f_1(x_1, \alpha)] + z_2 \left[\underbrace{\frac{\partial V_1}{\partial x_1} + \phi - \dot{\alpha}}_{=-z_2} \right] + z_2 z_3$$

$$\phi(x_1, x_2) = \dot{\alpha} - z_2 - \frac{\partial V_1}{\partial x_1}$$

$$z_3 = x_3 - \phi$$

$$\dot{z}_3 = \dot{x}_3 - \dot{\phi}$$

$$V_3(x_1, z_2, z_3) = V_2(x_1, z_2) + \frac{1}{2} z_3^2$$

$$\dot{V}_3 = \dot{V}_2 + z_3 \dot{z}_3$$

$$= \text{neg. def. (in } x_1, z_2) + z_2 z_3 + z_3 (u - \dot{\phi})$$

$$= \text{neg. def. (in } x_1, z_2) + z_3 \underbrace{(z_2 + u - \dot{\phi})}_{=-z_3}$$

$$u = -z_2 - z_3 + \dot{\phi}$$

Revisit Ex. $\dot{x}_1 = x_1^2 + x_2 = x_1^2 + \alpha + x_2 - \alpha$

$$= x_1^2 + \alpha + z_2$$

$$= f_1(x_1, \alpha) + z_2$$

$$\dot{x}_2 = u$$

$$\alpha(x_1) = -x_1^2 - x_1$$

$$V_1(x_1) = \frac{1}{2} x_1^2$$

$$z_2 = x_2 - \alpha(x_1)$$

$$\dot{z}_2 = u - \dot{\alpha}$$

$$V_c(x_1, z_2) = V_1(x_1) + \frac{1}{2} z_2^2$$

$$\dot{V}_c = \dot{V}_1(x_1) + z_2 \dot{z}_2$$

$$= \frac{\partial V_1}{\partial x_1} (f_1(x_1, \alpha) + z_2) + z_2 (u - \dot{\alpha})$$

$$= \underbrace{\frac{\partial V_1}{\partial x_1} f_1(x_1, \alpha)}_{-x_1^2} + z_2 \underbrace{\left(\frac{\partial V_1}{\partial x_1} + u - \dot{\alpha} \right)}_{= -z_2}$$

$$= -x_1^2 - z_2^2$$

$$\frac{\partial V_1}{\partial x_1} + u - \dot{\alpha} = -z_2$$

$$\Rightarrow u = \dot{\alpha} - z_2 + \frac{\partial V_1}{\partial x_1}$$

$$= -2x_1 \dot{x}_1 - \dot{x}_1 - x_2 + \alpha(x_1) + x_1$$

$$= \dots$$

$$\Downarrow$$

$$\begin{bmatrix} x_1 \\ z_2 \end{bmatrix} = 0 \text{ is GAS} \Rightarrow x_1 \rightarrow 0$$

$$z_2 \rightarrow 0$$

$$\Downarrow$$

$$x_2 - \alpha(x_1) \rightarrow 0$$

$$x_2 \rightarrow \underbrace{\alpha(x_1)}$$

$$\alpha(x_1) \rightarrow 0$$

$$x_2 \rightarrow 0$$

Alternatively,

$$\dot{x}_1 = f_1(x, \alpha) + z_2$$

$$z_2 \equiv 0 \rightarrow x_1 \rightarrow 0$$

$$z_2 \rightarrow 0$$

$$\Rightarrow x_1 \rightarrow 0$$

X

$$\dot{x}_1 = 0.5 x_1 + x_2$$

$$\dot{x}_2 = u$$

$$\alpha(x_1) = -0.5 x_1 - x_1$$

$$z_2 = x_2 - \alpha$$

$$\begin{bmatrix} x_1 \\ z_2 \end{bmatrix} \rightarrow 0$$

$$x_2 \rightarrow \alpha(0) = -1$$

Backstepping can be applied recursively to systems of the form:

$$\dot{x}_1 = f_1(x_1) + g_1(x_1) x_2$$

$$\dot{x}_2 = f_2(x_1, x_2) + g_2(x_1, x_2) x_3$$

$\vdots$

$$\dot{x}_n = f_n(x) + g_n(x) u$$

$$\dot{x}_{n+1} = f_n(x_1, \dots, x_n, x_{n+1})$$

Ex.

$$\dot{x}_1 = (x_1 x_2 - 1) x_1^3 + (x_1 x_2 + x_2^2 - 1) x_1$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = u$$

strict feedback form

not in strict feedback form

$$\dot{x}_1 = (x_1 x_2 - 1) x_1^3 + (x_1 x_2 - 1) x_1$$

$$= f_1(x_1) + g_1(x_1) x_2$$

$$= x_1^4 x_2 - x_1^3 + x_1^2 x_2 - x_1$$

$$= \underbrace{(-x_1^3 - x_1)}_{f_1} + \underbrace{(x_1^4 + x_1^2)}_{g_1} x_2$$

$$\dot{x}_1 = (x_1 x_2 - 1) x_1^3 + (x_1 x_2 - 1) x_1 + u$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_2$$

$$y_1 = x_2$$

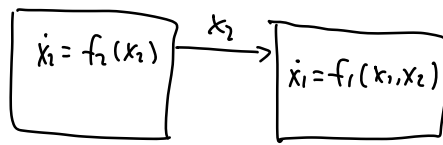
$$\dot{y}_1 = y_2$$

$$y_2 = x_3$$

$$\dot{y}_2 = y_3$$

$$y_3 = x_1$$

$$\dot{y}_3 = () u$$



$$x_1 \in \mathbb{R}^n$$

$$x_2 \in \mathbb{R}^m$$

• If $x_1 = 0$ is GAS for $\dot{x}_1 = f_1(x_1, 0)$

• If $x_2 = 0$ is GAS for $\dot{x}_2 = f_2(x_2)$

$\Downarrow$

$(x_1, x_2) = 0$ is GAS for the interconnection? X

Ex. $\begin{cases} \dot{x}_1 = -x_1 + x_1^2 x_2 \\ \dot{x}_2 = -x_2 \end{cases}$ exhibit finite escape time

— $\dot{x}_1 = -x_1 + x_1^2 u$ (u exp. decaying)

$$u = \varepsilon$$

$$\dot{x}_1 = -x_1 + x_1^2 \varepsilon$$

$$= x_1 (\varepsilon x_1 - 1)$$

$$\frac{dx}{x(\varepsilon x - 1)} = dt$$

$$dx \left[\frac{1}{\varepsilon x - 1} - \frac{1}{\varepsilon x} \right] = \frac{dt}{\varepsilon}$$

$$\ln \left| \frac{\varepsilon x - 1}{\varepsilon x} \right| \Big|_{x_0}^x - \ln \left| \frac{\varepsilon x}{\varepsilon x} \right| \Big|_{x_0}^x = \frac{t}{\varepsilon}$$

$$\frac{\varepsilon x - 1}{\varepsilon x_0 - 1} \frac{\varepsilon x_0}{\varepsilon x} = e^{\frac{t}{\varepsilon}}$$

$$\frac{\varepsilon x - 1}{\varepsilon x} = \frac{\varepsilon x_0 - 1}{\varepsilon x_0} e^{\frac{t}{\varepsilon}}$$

$$x_0 = \frac{1.1}{\varepsilon}$$

$$\frac{\varepsilon x - 1}{\varepsilon x} = \frac{0.1}{1.1} e^{\frac{t}{\varepsilon}} = \frac{1}{11} e^{\frac{t}{\varepsilon}}$$

$$\frac{1}{\varepsilon x} = \frac{e^{\frac{t}{\varepsilon}} - 11}{11}$$

$$\therefore x = -\frac{11}{\varepsilon(e^{\frac{t}{\varepsilon}} - 11)} = \frac{11}{\varepsilon(11 - e^{\frac{t}{\varepsilon}})}$$

finite escape time

Input-to-state Stability

The system $\dot{x} = f(x, u)$, $f(0,0)=0$, is said to be input-to-state (ISS) stable if:

$$|x(t)| \leq \beta(|x(0)|, t) + \gamma \left(\sup_{0 \leq \tau \leq t} |u(\tau)| \right)$$

where β is class KL function

γ is class K function

- If $u \equiv 0$

$$\Rightarrow |x(t)| \leq \beta(|x(0)|, t) \Rightarrow \begin{matrix} x=0 \\ \text{GAS} \end{matrix}$$

$$\text{ISS} \Rightarrow \text{GAS}$$

- If $u(t) \rightarrow 0$

$$x=0 \text{ is GAS}$$

Ex. $\dot{x} = -x^r + x^s u$

$$r: \text{odd integer} \Rightarrow \text{ISS}$$

$$r > s$$